

EXPLAINABLE MEDICAL AI DIAGNOSTIC REPORT

Automated Chest X-Ray Screening for Pneumonia Detection with Visual Explanations

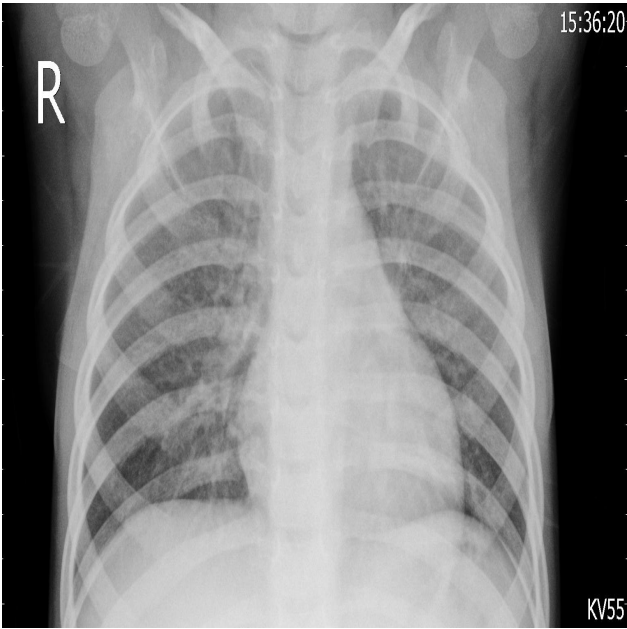
Patient Name:	Anonymous Patient	Report Date:	2026-07-01 01:48:15
Diagnostic ID:	REC-#5	Model Version:	PneumoniaCNN v1.0.0

AI DIAGNOSTIC RESULT: PNEUMONIA
Probabilities — Pneumonia: 50.72% | Normal: 49.28%

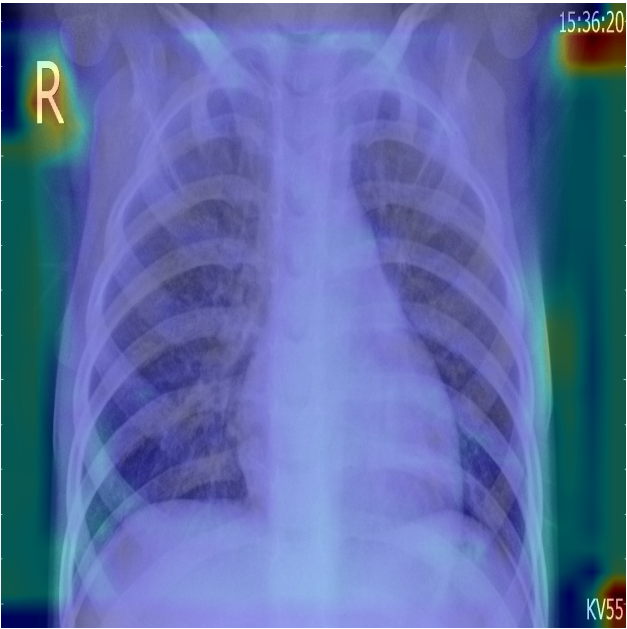
Confidence score: 50.72%
Analysis Processing Time: 6.04 seconds

IMAGE FINDINGS & EXPLAINABLE AI ANALYSIS

Original Chest X-Ray



Grad-CAM Localization Heatmap



AI DIAGNOSTIC FINDINGS CHECKLIST

- ✓ Lung Opacity: The model detected areas of increased density in pulmonary fields.
- ✓ Lower Lobe Abnormalities: Focal densities and consolidations localized in the lobes.
- ✓ High Density Infiltrates: Fluid density regions detected, strongly shifting probability values.

Visual Explanation Summary: The AI model identified localized regions of high density and opacity, highlighted by the red/yellow regions in the Grad-CAM heatmap above. These patterns correspond to consolidated lung parenchyma and fluid infiltrates, which are standard clinical indicators of pneumonia infection.

Disclaimer: This diagnostic report was generated by an experimental Artificial Intelligence system. It is intended solely for educational, portfolio demonstration, and decision-support purposes. This report does NOT constitute a medical diagnosis. The findings and explanations should always be verified by a board-certified physician or radiologist.